

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $C = ABA^T$ and $X = A^T C^2 A$, then $\det X$ is equal to :

- (1) 243
- (2) 729
- (3) 27
- (4) 891

Q2 - 2024 (01 Feb Shift 2)

Let $A = I_2 - MM^T$, where M is real matrix of order 2×1 such that the relation $M^T M = I_1$ holds. If λ is a real number such that the relation $AX = \lambda X$ holds for some non-zero real matrix X of order 2×1 , then the sum of squares of all possible values of λ is equal to :

Q3 - 2024 (27 Jan Shift 1)

Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Given below are two statements :

Statement I: $f(-x)$ is the inverse of the matrix $f(x)$.

Statement II: $f(x)f(y) = f(x+y)$.

In the light of the above statements, choose the correct answer from the options given below

- (1) Statement I is false but Statement II is true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Both Statement I and Statement II are true

Q4 - 2024 (27 Jan Shift 2)

Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A - xI| = 0$ be -1 and 3, then the sum of the diagonal elements of the matrix A^2 is.

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Q5 - 2024 (29 Jan Shift 1)

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ and $|2A|^3 = 2^{21}$ where $\alpha, \beta \in \mathbb{Z}$, Then a value of α is

- (1) 3
- (2) 5
- (3) 17
- (4) 9

Q6 - 2024 (29 Jan Shift 1)

Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2} A \left[(A + A^T)^2 + (A - A^T)^2 \right]$ is equal to

- (1) $A^2 + I$
- (2) $A^3 + I$
- (3) $A^2 + A^T$
- (4) $A^3 + A^T$

Q7 - 2024 (29 Jan Shift 2)

Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $|P^{-1}AP - 2I|$ is equal to

- (1) 26
- (2) 27
- (3) 66
- (4) 23

Q8 - 2024 (30 Jan Shift 2)

Let $R = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ be a non-zero 3×3 matrix, where $x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right)$

$\neq 0, \theta \in (0, 2\pi)$. For a square matrix M , let trace (M) denote the sum of all the diagonal entries of M. Then,

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among the statements:

(I) $\text{Trace}(R) = 0$

(II) If $\text{trace}(\text{adj}(\text{adj}(R))) = 0$, then R has exactly one non-zero entry.

(1) Both (I) and (II) are true

(2) Neither (I) nor (II) is true

(3) Only (II) is true

(4) Only (I) is true

Q9 - 2024 (31 Jan Shift 2)

Let A be a 3×3 matrix and $\det(A) = 2$. If

$$n = \det(\underbrace{\text{adj}(\text{adj}(\dots(\text{adj}(A))))}_{2024\text{--times}})$$

Then the remainder when n is divided by 9 is equal to _____.

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Answer Key

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Solutions

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Q1

$$A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix} \Rightarrow \det(A) = 3$$

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \det(B) = 1$$

$$\text{Now } C = ABA^T \Rightarrow \det(C) = (\det(A))^2 \times \det(B)$$

$$|C| = 9$$

$$\text{Now } |X| = |A^T C^2 A|$$

$$= |A^T| |C|^2 |A|$$

$$= |A|^2 |C|^2$$

$$= 9 \times 81$$

$$= 729$$

Q2

$$A = I_2 - 2MM^T$$

$$A^2 = (I_2 - 2MM^T)(I_2 - 2MM^T)$$

$$= I_2 - 2MM^T - 2MM^T + 4MM^T MM^T$$

$$= I_2 - 4MM^T + 4MM^T$$

$$= I_2$$

$$AX = \lambda X$$

$$A^2 X = \lambda AX$$

$$X = \lambda(\lambda X)$$

$$X = \lambda^2 X$$

$$X(\lambda^2 - 1) = 0$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

$$\text{Sum of square of all possible values} = 2$$

Q3

$$f(-x) = \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$f(x) \cdot f(-x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence statement- I is correct

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Solutions

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Now, checking statement II

$$f(y) = \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$f(x) \cdot f(y) = \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow f(x) \cdot f(y) = f(x+y)$$

Hence statement-II is also correct.

Q4

$$|A - xI| = 0$$

Roots are -1 and 3

$$\text{Sum of roots} = \text{tr}(A) = 2$$

$$\text{Product of roots} = |A| = -3$$

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{We have } a + d = 2$$

$$ad - bc = -3$$

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$$

$$\text{We need } a^2 + bc + bc + d^2$$

$$= a^2 + 2bc + d^2$$

$$= (a + d)^2 - 2ad + 2bc$$

$$= 4 - 2(ad - bc)$$

$$= 4 - 2(-3)$$

$$= 4 + 6$$

$$= 10$$

Q5

$$|A| = \alpha^2 - \beta^2$$

$$|2A|^3 = 2^{21} \Rightarrow |A| = 2^4$$

$$\alpha^2 - \beta^2 = 16$$

$$(\alpha + \beta)(\alpha - \beta) = 16 \Rightarrow \alpha = 4 \text{ or } 5$$

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Solutions

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Q6

$$AA^T = I = A^T A$$

On solving given expression, we get

$$\begin{aligned} \frac{1}{2} A \left[A^2 + (A^T)^2 + 2 A A^T + A^2 + (A^T)^2 - 2 A A A^T \right] \\ = A \left[A^2 + (A^T)^2 \right] = A^3 + A^T \end{aligned}$$

Q7

$$\begin{aligned} |P^{-1}AP - 2I| &= |P^{-1}AP - 2P^{-1}P| \\ &= |P^{-1}(A - 2I)P| \\ &= |P^{-1}| |A - 2I| |P| \\ &= |A - 2I| \\ &= \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69 \end{aligned}$$

So, Prime factor of 69 is 3 & 23

So, sum = 26

Q8

$$1. x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right) \neq 0$$

$$\Rightarrow x, y, z \neq 0$$

Also,

$$\sin \theta + \sin \left(\theta + \frac{2\pi}{3} \right) + \sin \left(\theta + \frac{4\pi}{3} \right) = 0 \forall \theta \in R$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\Rightarrow xy + yz + zx = 0$$

$$(i) \text{ Trace}(R) = x + y + z$$

$$\text{If } x + y + z = 0 \text{ and } xy + yz + zx = 0$$

$$\Rightarrow x = y = z = 0$$

Statement (i) is False

$$(ii) \text{ Adj}(\text{Adj}(R)) = |R|R$$

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Solutions

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$$\text{Trace}(\text{Adj}(\text{Adj}(\text{R})))$$

$$= xyz(x + y + z) \neq 0$$

Statement (ii) is also False

Q9

$$|A| = 2$$

$$\underbrace{\text{adj}(\text{adj}(\text{adj} \dots \text{adj}(a)))}_{2024 \text{ times}} = |A|^{(n-1)2024}$$

$$= |A|^{2024}$$

$$= 2^{2024}$$

$$2^{2024} = (2^2)^{2^{2022}} = 4(8)^{674} = 4(9-1)^{674}$$

$$\Rightarrow 2^{2024} \equiv 4 \pmod{9}$$

$$\Rightarrow 2^{2024} \equiv 9m + 4, m \leftarrow \text{even}$$

$$2^{9m+4} \equiv 16 \cdot (2^3)^{3m} \equiv 16 \pmod{9}$$

$$\equiv 7$$

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